

A Decluttering Lens for Scatterplots

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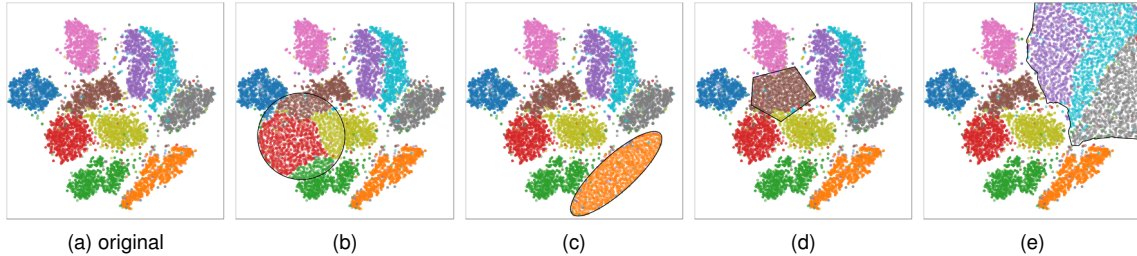


Fig. 1: Decluttering lenses on the MNIST dataset [14] with 10,000 samples, projected using UMAP [19]. In contrast to the classical magnifying lenses, these lenses allow for maximal decluttering of samples in the focus area. The shape of the lens and the regularization degree within the lens are interactively steered by the user depending on (c) the cluster shape, (d) distances to the neighboring clusters, or (e) available unoccupied space.

ABSTRACT

Scatterplots are among the most popular visualization methods for bivariate or multivariate data. While scatterplots scale well with the number of samples, visual clutter cannot be avoided with increasing data size. Density regularization is a common approach to declutter scatterplots. However, most existing density-equalizing algorithms are restricted to rectangular domains, which limits their applicability. In particular, they cannot operate within interactive lenses, a widely used approach for interactive data exploration. We present a numerical approach that generalizes density-equalizing transformations to domains of arbitrary shape, including concave regions. The definition of the regions of interest can be data-driven or interactive. We demonstrate the effectiveness of our method by implementing adaptive and flexible interactive lenses for enhanced data exploration in scatterplots, showcasing its versatility and potential for broader application.

Index Terms: Clutter reduction, scatterplots, density equalization, virtual lenses.

1 INTRODUCTION

Scatterplots are a widely used visualization tool for exploring relationships between two numerical variables. They enable the detection of trends, clusters, outliers, and facilitate correlation and regression analysis. Scatterplots are also the standard visualization for projections of multivariate data into 2D embeddings. However, as the number of data points increases, occlusion becomes a significant obstacle in visual data exploration and analysis. Over the decades, numerous approaches have been proposed to mitigate scatterplot overdraw [6]. Common techniques include data manipulation (e.g., aggregation [23] or down-sampling [4]), appearance adjustments (e.g., modifying point size and transparency [20]), and spatial transformations (e.g., point remapping [27]). Many of these

techniques can be combined and applied interactively under user control.

Spatial transformations for occlusion reduction have the desired property that they still represent all data items. They achieve that by either allocating more visual space to areas of interest or globally redistributing data points to optimize screen space usage [15]. The former includes zooming techniques and virtual lenses [32], while the latter encompasses regularization methods such as pixel remapping and rubber-sheet distortions. Each approach has its trade-offs: virtual lenses preserve global context but may still suffer from visual clutter in focused regions, whereas global spatial deformations improve point distinguishability but can distort the overall data distribution. In this work, we propose a local occlusion reduction method that combines the benefits of the *focus + context* paradigm with user-controlled regularization in the focal area.

To the best of our knowledge, most existing deformation-based decluttering techniques are designed for global application, predominantly in rectangular domains. Localizing these methods to specific regions of interest is often non-trivial or infeasible. Our work addresses this limitation by introducing a regularization approach that operates within arbitrarily shaped contours. Specifically, we adapt the state-of-the-art decluttering technique by Rave et al. [26] to enable localized sample redistribution.

Having defined a local density-equalizing regularization approach that operates on any shape, different application scenarios can be investigated. Common disk-shaped virtual lenses can be implemented, as well as virtual lenses of any user-defined shape. The virtual lenses' shape could also be data-driven, such as being defined by the data cluster shape, relative position with respect to other data structures, and available unoccupied screen space.

The key contributions of our work include:

- Modification of Rave et al.'s regularization method [26] for disk-shaped domains, eliminating artifacts caused by boundary singularities (e.g., corners), which was the main drawback of their work.
- Generalization of the density-equalizing regularization technique to arbitrary-shaped domains, including concave regions.

2 RELATED WORK

Scatterplots have a long application history [8] with extensive research on quality metrics and characterization. Behrisch et al. [1]

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systematically studied quality metrics, building on earlier work by Tukey and Tukey [33] who proposed *scagnostics* measures. These were refined through graph-theoretic approaches by Wilkinson et al. [35], while Matute et al. developed skeleton-based [18] and principal graph-based [17] summarization techniques. Sarikaya and Gleicher [29] provided a comprehensive survey of scatterplot designs organized by analysis tasks. For higher-dimensional data, scatterplot matrices [34] and dimensionality reduction techniques using scatterplots for visualizing 2D embeddings [31] have become standard tools, with Elmqvist et al. [7] and Sanftmann and Weiskopf [28] developing navigation methods for these multidimensional visualizations. Despite the wide usage of scatterplots and their long history, occlusion is still an issue for an increasing number of data points. Our goal is to make scatterplots more accessible within interactive scenarios by defining decluttering lenses that reduce occlusion within the focus area.

Virtual Lenses implement the *focus+context* paradigm through localized deformations. Early fisheye views by Furnas [9] established the foundation for smooth detail interpolation. Modern variants like *Jelly Lenses* [24] adapt their shape to data characteristics, while Molchanov and Linsen [21] developed arbitrary-shaped lenses for geospatial data. Tominski et al. [32] specifically addressed scatterplot applications, providing a taxonomy of lens techniques. Our approach also strives for virtual lenses of arbitrary shapes. However, while the methods listed above primarily focus on zooming operations, magnifying lenses, or changing the visual encoding / rendering style within the virtual lenses, our goal is to define a regularization within the virtual lenses to locally declutter the scatterplot based on equalizing point density. Since our decluttering approach is based on spatial transformation equalizing density distributions, data points that are very close to each other will still become clearly separated, which is barely the case for magnifying lenses. Thus, the available virtual lens space is more effectively used by our regularization approach.

Decluttering and Gridifying Algorithms usually aim at global reduction of overplotting and overlaps in scatterplots. Keim et al. [13] balanced overlap and distortion in generalized scatterplots, while Raidou et al. [25] employed pixel-based space-filling transformations. Grid-based approaches like those by Cutura et al. [2, 3] (using space-filling curves) and Hilaraca et al. [12] (solving glyph overlaps) achieve $O(n \log n)$ complexity, where n is the number of samples. Both pixel maps and space-filling curves are applied to rectangular domains. The integral image method by Rave et al. [27] achieves $O(n)$ complexity operations and allows for GPU acceleration, but is still limited to rectangular domains. In a follow-up paper, Rave et al. [26] improved the uniformity of the resulting distribution by increasing the angular resolution of the method. Our work generalizes these approaches by supporting arbitrary regularization domains, which allows their use within the context of virtual lenses of any shape.

Density-Equalizing Transformations are a broad class of methods, which the decluttering algorithms by Rave et al. [26, 27] represent a specific case of. Density equalization is the core operation for the construction of contiguous cartograms. Gastner and Newman's physics-inspired methods [10, 11] utilize diffusion and liquid-flow models for designing the transformations. Further, density equalization is intensively used in geometry processing, e.g., for mesh parameterization. Recent advances include *Boundary First Flattening* [30] for disk conformal maps, *Bijective Density-Equalizing Quasiconformal Maps* [16] for multiply-connected domains, and *Instant Transport Maps* [22]. These methods provide mathematical guarantees but often require expensive pre-computation or specialized domain constraints. Our sector-based approach bridges the gap between these rigorous techniques and practical visualization needs by offering real-time performance with quality comparable to Rave et al.'s methods.

3 BACKGROUND

Scatterplot decluttering via spatial transformation is an effective strategy to mitigate occlusion while preserving the global structure of the data. Recent work by Rave et al. [27] introduced a fast regularization method using *integral images* (InIm) to redistribute samples in an iterative optimization procedure. Their approach partitions the visual domain into four quadrants (or eight sectors when including tilted InIm) and computes displacements by comparing the local sample density to a uniform distribution. While efficient, this low angular resolution limits the uniformity of the resulting layout. The subsequent *sector-based transformation* [26] generalizes this idea by increasing the number of sectors around each sample, significantly improving spatial uniformity.

The sector-based regularization algorithm for a set of n samples in a 2D rectangular domain $\mathcal{S}$ operates as follows:

- **Sector definition:** For each sample $\mathbf{x}_u = (x_u, y_u)$, $u = 1, \dots, n$, the domain is partitioned into m equal-angle sectors $\mathcal{S}_l(\mathbf{x}_u)$, $l = 1, \dots, m$, centered at $\mathbf{x}_u$.
- **Anchor points:** Each sector's centerline intersects the domain boundary at two points, one within the segment and one outside the segment. The intersection point outside the sector is chosen as an *anchor point* $\mathbf{q}_l(\mathbf{x}_u)$.
- **Density evaluation:** The number of samples $s_l(\mathbf{x}_u)$ in each sector $\mathcal{S}_l(\mathbf{x}_u)$ is counted and compared to the expected uniform density $|\mathcal{S}_l(\mathbf{x}_u)|/|\mathcal{S}|$, where $|\mathcal{S}_l(\mathbf{x}_u)|$ denotes the area of sector $\mathcal{S}_l(\mathbf{x}_u)$ and $|\mathcal{S}|$ the total domain area.
- **Displacement computation:** The displacement vector for $\mathbf{x}_u$ is computed as

$$\mathbf{t}(\mathbf{x}_u) = \sum_{l=1}^m \mathbf{q}_l(\mathbf{x}_u) \cdot (s_l(\mathbf{x}_u)/n - |\mathcal{S}_l(\mathbf{x}_u)|/|\mathcal{S}|). \quad (1)$$

- **Iteration:** The samples are moved iteratively until quasi-equilibrium is reached, achieving a near-uniform distribution.

4 METHODS AND RESULTS

Both the InIm-based and sector-based regularizations by Rave et al. [26, 27] face several challenges and limitations related to the regularization domain shape. First, the rectangular domains introduce boundary artifacts, particularly near corners, where singularities distort nearby sample distributions. Second, the methods cannot be applied to non-rectangular domains. The latter is an intrinsic limitation for many existing regularization algorithms, including methods by Gastner et al. [11] and Nader and Guennebaud [22]. Other approaches (e.g., space-filling curves used by Cutura et al. [2, 3]) would require a significant modification to make them applicable to arbitrary-shaped domains.

In this paper, we address these limitations by first replacing the rectangular domain with a disk-shaped region to eliminate boundary artifacts (see Sec. 4.1). We then generalize the approach to arbitrary-shaped (including concave) domains, enabling localized regularization within arbitrary-shaped areas (see Sec. 4.3). This generalization to arbitrary shapes allows for the application of our approach to the concept of virtual lenses, where the lens domain that is used for regularization can be any user-defined region (including the commonly used disk shape) as well as a data-driven definition or derivation. Our implementation of the proposed algorithm will be made publicly available upon publication.

4.1 Regularization over Disk

Existing regularization methods, including both the InIm and sector-based approaches, operate on the entire scatterplot domain and aim to achieve uniform sample distribution across a rectangular area. While efficient for GPU implementation — particularly the

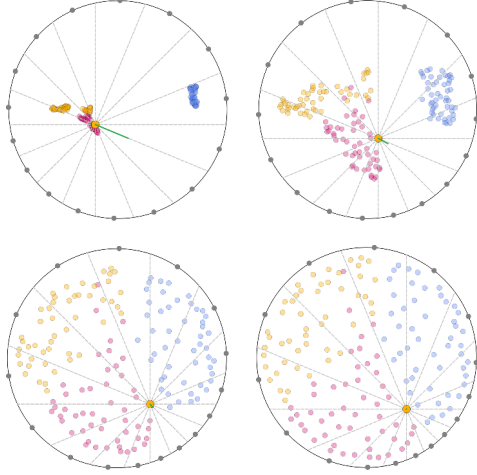


Fig. 2: Modified sector-based regularization approach for a disk-shaped domain. Top left: Initial distribution of samples within the domain. Sectors and their corresponding anchor points are shown for a single data sample. The sample's displacement during the first iteration is indicated by a green line. Subsequent images show the progressive regularization of samples through iterative application of the algorithm.

InIm-based method, which relies on parallel operations over rasterized textures — these methods exhibit notable uniformity degradation near domain boundaries, with the most severe artifacts occurring near the four corners. One potential solution would be to apply corrective measures, such as local nudging, in these problematic regions. However, in this work, we propose a more fundamental approach: changing the geometry of the regularization domain itself by adopting a disk-shaped area, thereby eliminating boundary singularities entirely and conceptually.

The transition to a disk-shaped domain requires two key modifications to the sector-based algorithm:

- **Anchor point redefinition:** For each sample $\mathbf{x}_u$, $u = 1, \dots, n$, the anchor points $\mathbf{q}_l(\mathbf{x}_u)$ for sectors $\mathcal{S}_l(\mathbf{x}_u)$, $l = 1, \dots, m$, are now defined as the intersections between the sectors' centerlines and the bounding circle of the disk (rather than the rectangle edges). Again, the intersection point outside the sector is the one that is used as the anchor point $\mathbf{q}_l(\mathbf{x}_u)$.
- **Sector area computation:** The area $|\mathcal{S}_l(\mathbf{x}_u)|$ of each sector of sample $\mathbf{x}_u$ is recomputed as the area of the segment bounded by the sector's radial lines and the disk boundary. This area is simply the area of the triangle $\Delta \mathbf{x}_u \mathbf{s}_l \mathbf{s}_{l+1}$ plus the area of the enclosed circular segment.

With these modifications, the displacements $\mathbf{t}(\mathbf{x}_u)$ can be computed as defined in Equation 1, and the iterations can be executed. These modifications preserve all desirable properties of the sector-based approach — including smooth displacement fields and neighborhood preservation — while removing corner-induced artifacts, see Fig. 2. Notably, the disk-shaped domain also serves as the foundation for our generalized arbitrary-domain regularization, enabling new applications like focus+context visualization through virtual lenses.

4.2 Circular Virtual Lens

Traditional virtual lenses typically employ disk-shaped lens domains, as the circular symmetry enables simple distortion formulae that smoothly and often isotropically magnify the lens interior while preserving geometric relationships. Our approach differs funda-

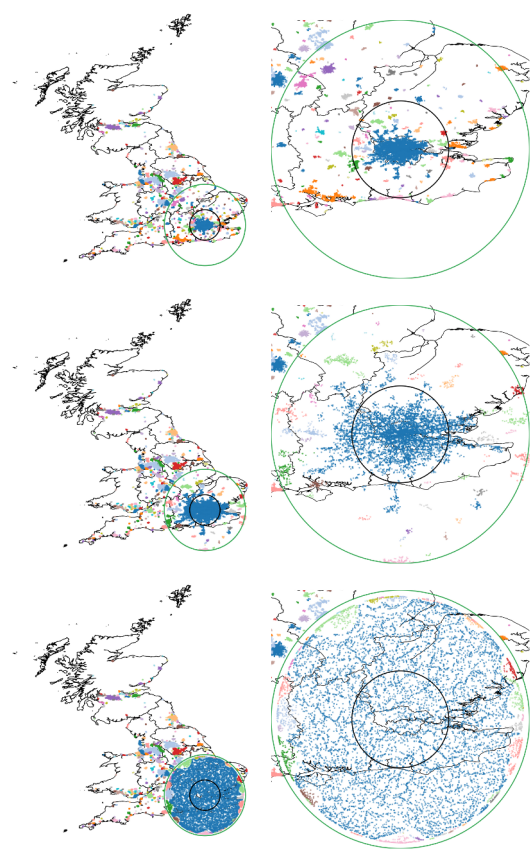


Fig. 3: Regularizing virtual lens applied to the *UrbanGB* dataset. Boundaries of the selection and regularization domains, D_1 and D_2 , are shown as black and green circles, respectively. The user can interactively change the size of the regularization area (green circle), adaptively allocating the screen space for the lens (left and right columns). Different rows demonstrate varying degree of the samples' regularization, smoothly transitioning from original samples' distribution (upper row) to fully regularized layout (lower row).

mentally by *regularizing* the data within the lens domain rather than merely magnifying it. This data-driven strategy optimizes space utilization and improves sample discrimination while maintaining smooth transitions between focus and context regions.

We performed tests using the *UrbanGB* dataset from the UCI machine learning repository [5], which contains 360,177 instances with two attributes (longitude and latitude) of road accidents that occurred in urban areas in Great Britain. We downsampled the original data by randomly selecting 10% of the instances. The results are presented in Fig. 3. The user selects a region of interest by locating the inner circle of the lens shown in black on the map. When the lens is activated, the samples in the selection area D_1 are regularized over the disk D_2 outlined in green. The user can interactively change the radius of the regularization disk for finer detailization. Additionally, the regularization degree can be smoothly changed interactively by controlling the number of regularization iterations. Thus, the proposed virtual lens can just magnify the original data structures in focus (no regularization) or fully regularize the selected samples for accessing and inspecting them individually. User-controlled smooth transition between these two modes allows the user to find an optimal trade-off between structure preservation and data decluttering, as demonstrated in the accompanying video.

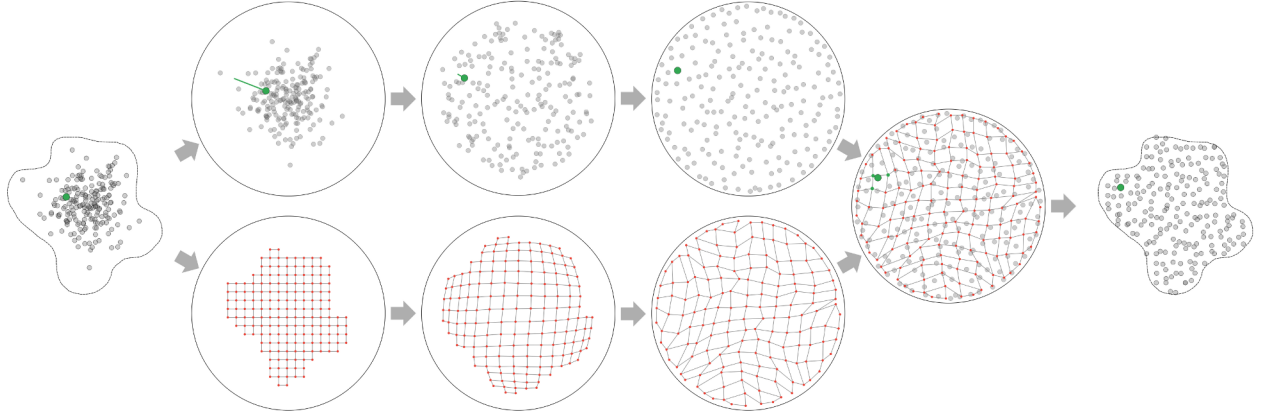


Fig. 4: Flow diagram of the algorithm regularizing samples over arbitrary-shaped domain. The selected data samples and the boundary of the regularization domain $D_1 = D_2$ are shown in the first plot. The upper path illustrates the samples' regularization over an auxiliary disk-shaped domain. The lower path shows the regularization of a regular grid in D_2 over the same disk. The results of both paths are combined and barycentric coordinates of samples are computed. The final result is obtained by reversing the grid regularization (lower path) and computing the samples positions in D_2 using the pre-computed barycentric coordinates.

4.3 Generalization to Arbitrary-Shaped Domains

While for simplicity a disk-shaped virtual lens is common, the lens shape does not need to be circular. Instead, it may be beneficial to adapt to the available empty space or to user-defined shapes. A naive generalization would place anchor points along boundaries of an arbitrary-shaped lens, but our tests demonstrate that this often pushes samples outside the domain. Artificially increasing external density to repel samples proved equally ineffective. We, instead, propose a two-stage transformation for regularizing arbitrary-shaped domains by using auxiliary mappings to a disk. Figure 4 illustrates the work flow. The method operates with two key domains: $D_1 \subseteq \mathbb{R}^2$ defining the selection domain within the scatterplot (containing samples to regularize) and $D_2 \subseteq \mathbb{R}^2$ representing the lens domain (the regularization target area). For virtual lenses, typically, $D_1 \subseteq D_2$, which we assume in the following for the simplicity of the exposition.

In the first stage, we construct two density-equalizing transformations over the same disk. For the first transformation, we regularize the selected samples (upper path in Fig. 4). For the second transformation, we generate an auxiliary regular grid in D_2 and regularize the grid nodes over the disk (lower path in Fig. 4).

In the second stage, we combine the results of both transformations by overlaying the regularized samples and the (triangularized) grid in the disk-shaped domain. For each regularized data sample, we find the triangle of connected grid nodes that overlaps with the sample. We compute the barycentric coordinates of the sample relative to these nodes. We then invert the grid regularization by effectively following the lower path shown in Fig. 4 in reversed order. Finally, we compute the positions of the data samples in D_2 using the previously stored barycentric coordinates.

In some cases, especially if the grid size is chosen too large, the overlapping triangle does not exist. Then, we compute the clamped barycentric coordinates with respect to the closest triangle. So, we avoid boundary-crossing of the samples after inverting the grid regularization. A sufficiently small grid size alleviates this issue.

4.4 Virtual Lenses of Arbitrary Shape

Since the shape of the proposed decluttering lens can be defined arbitrarily, it can be interactively adapted to suit the context and the structure of the data being explored. Figure 1 presents a series of snapshots illustrating the exploration of the MNIST dataset projected using UMAP. The original projection in Fig. 1a reveals

several clusters with varying degrees of sample occlusion. A disk-shaped lens is shown in Fig. 1b. In the next exploration step, the user customizes the lens shape to match the form of a particular cluster (Fig. 1c), which reduces regularization distortions and better preserves the sample order. Figures 1d and 1e further demonstrate how the lens shape can be adapted based on the relative positions of clusters and the available empty space near the domain boundaries. All illustrated lenses apply maximum regularization to the selected samples, providing an optimal level of decluttering. Since the degree of regularization is controlled interactively by the user, the proposed lenses offer a highly flexible tool for data exploration. The accompanying video presents additional examples of real-time applications using the proposed decluttering lens.

5 DISCUSSION AND CONCLUSION

In this work, we presented a spatially adaptable regularization algorithm that enables localized density regularization for focused data decluttering. Our method addresses artifacts associated with the sector-based regularization near domain boundaries. We extended the original approach to disk-shaped domains with proper modifications. Furthermore, we employed disk-based regularization as an intermediate step toward supporting arbitrary-shaped domains. For non-convex target domains, we prevented cross-boundary sample mappings using accurate barycentric interpolation.

Our method supports interactive control over both the degree of applied regularization and the granularity of the auxiliary grid. The former parameter allows for a smooth trade-off between structure preservation and clutter reduction within the lens. The latter should be selected based on the geometry and size of the regularization domain. A detailed numerical evaluation of the effects of grid size is left for future work.

Flexibility in domain shape enables further advancement of regularization-based techniques and broadens their range of applications. In particular, density equalization can now be applied either separately or progressively across different spatial scales, opening up new possibilities such as the construction of hierarchical contiguous cartograms. Similar to the JellyLens, the proposed virtual lens can adapt its shape to the local geographical context. In the current implementation, shape fitting should be performed manually by the user. However, an automatic approach (based, for instance, on clustering methods in the 2D visual domain) is feasible and could be developed in future work.

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